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A First Course in College Calculus

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To the Instructor

Introduction

This is a set of problems that was used for an introductory analysis course at Azim Premji University in Bangalore, India. This course was taken by mathematics majors in their second semester, following a course in discrete mathematics, also taught using IBL.

The first two offerings of this course, in 2019 and 2020, used modified versions of WT Ingram's *Foundations of Calculus* [1]. While the perspective and broad approach fit in perfectly with our curricular needs, we decided to try a different approach for the next iteration that addressed some challenges faced in the classroom.

The first issue we wanted to address was that most students join our programme after having done a lot of computational work, and expecting college maths to consist of similar but harder computations. In particular, most of them have studied various techniques for differentiation and integration, but very few schools go into the meaning of these processes. Before plunging into an axiomatic universe, we wanted students to work with examples and be convinced of the need for formal language and rigorous arguments.

Second, while our classes are taught in English, quite a few of our students have studied in a different medium of instruction in school. Others come from systems where the medium of instruction is officially English, but teaching and discussions happen in a local language. The course notes we originally used avoided stating axioms in formal mathematical language at the beginning, which was very helpful for those comfortable with English. However, students for whom English was a second or third language were at quite a disadvantage trying to parse long sentences and understand the (important) role being played by each preposition. So we wanted language to be less important at the beginning of the course.

These motivated us to develop an IBL calculus course that drew on existing courses in spirit, but which begins with concrete examples and computational questions, through which formal definitions and axioms (and their necessity) are motivated.

[1] Journal of Inquiry-Based Learning in Mathematics, No. 11, (Feb. 2009)

Structure and flow of the problem sets

Everything in this course is done with sequences – completeness of the reals, continuity of functions, integration, and differentiation are all introduced in the language of sequences. There are two main reasons for this: first, the definition of convergence of a sequence takes time to process, and there seems to be an advantage in having the same definition used over and over again through the course. Second, approximations and computations can be done to process the definition and build up intuition for it.

We describe how a course following these worksheets could unfold. The end-goal is to prove the Intermediate Value Theorem. If a class works fast, there is no shortage of further results they may hypothesise or prove. But, if a class moves slowly, there are specific topics that can be shortened or converted into in-class exercises or interactive lectures. These worksheets are meant for students who have already encountered the definition of a function, and have some familiarity with logic. Depending on the background of the students, these can be introduced in a couple of lectures at the beginning, or reviewed quickly when needed.

The course begins with an interactive lecture on what students think calculus is. After we review motivational questions about estimating the area under a curve and rate of change of a function, we observe that these both involve limiting processes. (This can be done even if many students in a class have not done calculus before.) Some students may remember the fundamental theorems of calculus, and few may have seen Rolle's Theorem (pictorially, not formally). The class is then asked to reflect on the motivation in using functions on the reals, especially if someone's aim is to apply these results to a real-life problem. We look (informally) at a few examples of functions on the rationals, where key results in calculus fail. Thus the aim of the course is stated as trying to better understand the limiting process, and the role played by the structure of the reals in the results of calculus.

The worksheets start with simplistic versions of the upper and lower Riemann sums (using subintervals of equal length and maximum/minimum values instead of suprema and infima). Students work through explicit calculations in a few easy cases, and also encounter a step function and the Dirichlet function. The examples highlight the limiting process in integration. After a few more computational exercises that motivate the definition of convergence of a sequence, we move on to a formal definition of convergence and integrability, and verify whether or not each of the previous examples was integrable. Students can pause at this stage and reflect on the shortcomings of the definition of integrability given here.

Differentiability is treated similarly, with the derivative being defined through sequences. Since the approach is similar to that taken for integration, less scaffolding is provided to the students, and they are asked to come up with a definition of differentiability through some motivational examples.

The class then moves on to studying sequences, the tools through which differentation and integration were defined. They observe patterns in sequences on the reals, only some of which hold for sequences on the rationals. This leads to an observation of a major structural difference between the reals and the rationals, namely that the bounded monotonic sequences always converge in the former, but may not in the latter.

In worksheets 1-5, students will probably make use of various results that they may not be able to justify. These may include various rules around manipulating inequalities, the Archimedean property, the well-definedness of the ceiling and floor functions, and the density of the rationals. The instructor should highlight these during presentations and keep note of them somewhere.

In the following worksheet (Sheet 6), we start with axioms for the reals. It may be useful to have another lecture to smoothen this transition with a reminder of the aims of the course, and a discussion on what we want from a set of axioms. Completeness of the reals is given by an axiom that bounded monotonic sequences converge. The worksheet guides students through proving various consequences of completeness, including the ones stated in the paragraph above. The instructor can modify this worksheet to include other results that their students have made use of in previous worksheets. At the end of this worksheet, students should be convinced that every result they have proven so far follows from the axioms.

Sheets 7 and 8 introduce continuity (also through sequences), and the course can end with a proof of the intermediate value theorem. The proof technique presented in Sheet 8 can be modified based on the attempts made by the students at the end of Sheet 7.

The final worksheet, Sheet 9, has some additional questions if there is time, or if someone wants to go further after the course ends.

Assessments

Assessment consisted of weekly assignments, a midterm test, a final test and a discussion/participation grade. Following the grading structure in various IBL courses, the discussion grade each week was between 0 and 4 out of 4 marks, with $2,3,4$ for making a presentation of varying correctness and completeness, and 1 for not presenting, but joining in the discussion.

The worksheets were supplemented by practice problem sets. These usually consisted of a few computational questions, a few short proofs and one or two optional harder questions. Weekly homework assignments consisted of just a couple of questions, either directly from the worksheets or from the practice problems.

Twice during the semester, students were asked to explain the bigger ideas of the course (like lower and upper sums and the role played by sequences in the definition of integrability) through short write-ups, aimed at a high school student. These provided students an opportunity to consolidate and articulate their understanding. Write-ups of this form can also give the instructor some glimpses into student understanding of the big-picture ideas.

Instructor experience and suggestions for using these worksheets

These problem sets were used in 2021-22 and 2022-23, first for a class of 35 and then for a class of 22. Both classes consisted of first year students majoring in mathematics or a enrolled in a joint degree in mathematics and education.

The chapters are divided into worksheets, each of which can be handed out to students just as they are finishing up the previous one. For Sheet 5, it would be good to hand out Part 1 first, see what ideas students come up with while filling in the table in Problem 51, and then hand out Part 2, or a modified version of Part 2 that factors in aspects of the discussion from the first part.

Sheet 4, which introduces the idea of differentiability through sequences, can be converted to an in-class activity or an interactive lecture if the class is going slowly.

Sheet 7 ends with a statement of the intermediate value theorem. Students are unlikely to come up with a complete proof, but may have some ideas. Sheet 8 can be modified according to these ideas.

The worksheets assume students have seen some programming (in particular, Python). Questions involving Python can be replaced by some other programming language or by asking students to use a calculator or spreadsheet to make calculations, and to write a step-by-step algorithm for problems that ask for code. The questions involving sequences and algorithms are to build intuition both for sentences involving quantifiers, and for the definition of convergence (at least for monotonic sequences).

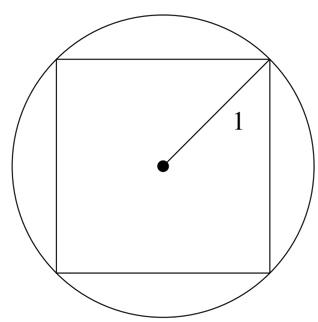
To the Student

What these problem sets are about

The aim of these worksheets is to give you a glimpse into the machinery behind calculus. Most students who have worked through these problems have already seen some calculus in high school, but not having done so shouldn't stop you from working through.

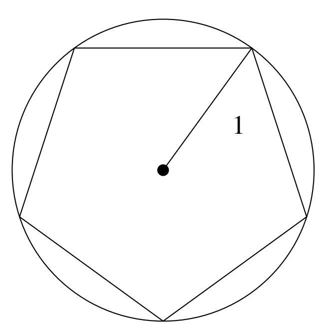
You have seen that the area of a circle of radius 1 is π. Here's one way to arrive at this result:

Draw a square with vertices on the circle that also shares a centre with the circle, as below:



Calculate the area of the square, and note down the value you get. Whatever value you get is less than the area of the circle.

Now draw a regular pentagon in the same circle, as below:



Calculate the area of the pentagon (for example, by dividing the pentagon into five congruent triangles and using some trigonometry). Again, note down the value you get, and observe that the area of the pentagon is still less than that of the circle, but larger than that of the square.

Continue with regular hexagons, heptagons, octagons, and so on. As soon as you are ready to, calculate the area of a regular n-gon with vertices on the circle and whose centre is the centre of the circle.

Each time we calculate the area of the n-gon, we obtain a number that is less than the area of the circle. But each time we get a little closer to the area of the circle. So we could say that the areas of the polygons "tend to" the area of the circle. And if you calculate the areas of the n-gons for some large number of natural numbers, you could easily believe the the values do indeed tend to π.

But maybe this pattern holds for the first one million numbers, and then suddenly changes. Or maybe these numbers look like they are getting closer together, but they don't actually get closer to any particular value. Maybe they get closer to two values, π and something else. Maybe if I try and get closer to the area with n-gons and you try and get closer to the area with rectangles, we'll tend to different values. And, anyway, what do we really mean by "tending to"?

This set of problems starts with questions (that you may have seen before), such as how to find the area under a graph or how to find the tangent line at a point on a graph of a function, which lead to the same questions as in the paragraph above. The problems guide you through one approach towards answering these questions, and also through some further observations and hypotheses that arise during the process.

How to approach these problems

The first worksheet starts with a formal definition. If you can't understand the definition, don't panic! The problems that follow are there to help unravel the definition, so work through them slowly and keep pausing to think about what you are doing and what the definition is getting at. As you work on your own, sit through other people's presentations, and share your own calculations and insights with your classmates, the definition should become clearer.

This is the idea behind each worksheet. The aim is not to master the definitions at the beginning and use your mastery to solve all the problems. Your aim is to understand the meaning and implications of the concepts being defined, and the problems are a way to guide you through this process. Think of the problems in the worksheets as conversation starters. They provide a common vocabulary to help you frame your ideas and talk about them, and some directions to explore. But you'll get the most out of them if you start playing with your own examples, and coming up with your own questions and hypotheses.

As the worksheets progress, there are fewer examples and computations, and more proofs. Some of the proofs involve carefully opening up definitions, but a few need more time and creativity (and some luck). If you find the definitions or theorem statements hard to parse, try playing around with some concrete examples. You can also try proving simpler versions of the results or special cases.

Prerequisites

These problems are ideally done as part of a course, where your instructor will fill in any prerequisite knowledge you need and provide big-picture ideas and summaries as you progress. They are also best done with other people.

If you want to go through them on your own, then you should ideally have done some calculus in school, know a little about sets and be familiar with the notation for common subsets of the real numbers, know what a function is and be familiar with the graphs of some standard functions on the reals, and have done some programming.

The following notations are used in these notes:

- $\mathbb{N}$ denotes the set of all natural numbers $\{1, 2, 3, \ldots\}$
- $\mathbb{Z}$ denotes the set of all integers
- $\mathbb{Q}$ denotes the set of all rational numbers
- $\mathbb{R}$ denotes the set of all real numbers

Chapter 1

Preliminary explorations with areas and tangents

Sheet 1

Definition 1. *Given a function $f : [a,b] \to \mathbb{R}$, divide the domain $[a,b]$ into n closed subintervals of equal length l. For each $1 \leq k \leq n$, let f^k be the largest value (if it exists) reached by the function on the k^{th} subinterval.*

*The **upper** n-**area** of f is defined as*

$$U_n(f) := (f^1 + f^2 + \cdots + f^n) \times l.$$

Example 2. *Divide the interval $[-3,5]$ into ten subintervals of equal length. What is the length of each subinterval? Divide the interval $[a,b]$ into n subintervals of equal length. What is the length of each subinterval?*

Example 3. *Sketch the graph of the function $f : [0,1] \to \mathbb{R}; f(x) = 7$. Divide $[0,1]$ into two pieces. What are f^1 and f^2? Use this to calculate $U_2(f)$. Similarly, calculate $U_5(f)$. Then calculate $U_n(f)$ for a general natural number n. Give a geometric meaning to $U_n(f)$.*

Example 4. *Let $g : [a,b] \to \mathbb{R}$ be given by $g(x) = c$ for some fixed real number c. What does the graph of g look like? Calculate $U_n(g)$, and explain its geometric meaning.*

Problem 5. *Come up with a definition of the **lower** n-**area** of a function $f : [a,b] \to \mathbb{R}$. We will denote this by $L_n(f)$. Explain the geometric meaning of $U_n(f)$ and $L_n(f)$.*

Example 6. *Calculate $L_n(g)$ for the map in Example 4 above.*

Example 7. *Consider $f : [0,1] \to \mathbb{R}; f(x) = x$. Find $U_n(f)$ and $L_n(f)$. What is $U_n(f) - L_n(f)$?*

Problem 8. *For the function in the example above, use Python to calculate $U_n(f)$, $L_n(f)$, and $U_n(f) - L_n(f)$ for $1 \leq n \leq 100$. What do you observe?*

Write a program that keeps calculating the difference $U_n(f) - L_n(f)$ *for larger and larger* n *until it is smaller than* 3×10^{-4}.

Example 9. *We now look at the function* $f : [a,b] \to \mathbb{R}; f(x) = x$. *Calculate* $U_n(f), L_n(f)$, *and* $U_n(f) - L_n(f)$. *Explain what these are geometrically. What happens to them as* n *gets larger?*

Example 10. *Consider* $f : [0,1] \to \mathbb{R}; f(x) = x^2$. *Calculate* $U_n(f)$ *and* $L_n(f)$.

What do you observe about $U_n(f), L_n(f)$ *and* $U_n(f) - L_n(f)$ *as* n *gets larger? Explain the meaning of your observations using the graph of* f.

Example 11. *Consider* $g : [-1,1] \to \mathbb{R}$ *given by*

$$g(x) = \begin{cases} 1 & \text{if } -1 \leq x \leq 0 \\ 2 & \text{if } 0 < x \leq 1 \end{cases}.$$

Sketch g, *and fill in the following table:*

n	1	2	3	4	5	6	7	8
$U_n(g)$								
$L_n(g)$								

What do you observe?

Problem 12. *Find a general expression for* $U_n(g)$ *and* $L_n(g)$ *in Example 11 above. What do you observe about* $U_n(g) - L_n(g)$ *as* n *increases?*

Example 13. *Consider* $h : [0,1] \to \mathbb{R}$ *given by*

$$h(x) = \begin{cases} 1 & \text{if } x \text{ is a rational number} \\ 0 & \text{if } x \text{ is an irrational number} \end{cases}$$

Find $U_n(h)$ *and* $L_n(h)$. *What do you observe about* $U_n(h) - L_n(h)$ *as* n *increases?*

Sheet 2

Definition 14. *Let A be a set. A* ***sequence in*** *A is a function $f : \mathbb{N} \to A$. We denote a sequence by $(f(n))_{n=1}^{\infty}$ or $(a_n)_{n=1}^{\infty}$, where $a_n = f(n)$. We often just drop the index and write (a_n) or $(a_1, a_2, a_3, \dots)$.*

Definition 15. *Given a sequence (a_n), and a natural number k, the* ***k-tail*** *of the sequence is the new sequence $(a_k, a_{k+1}, a_{k+2}, \dots)$.*

Example 16. *Draw the number line. Mark all the real numbers whose distance from -3 is less than 7. On a different number line, mark all the real numbers whose distance from 1.2 is greater than 0.21.*

Problem 17. *Suppose a, b, D are real numbers, $D > 0$. Come up with a mathematical expression that ensures that the distance between b and a is less than D. When this happens we often say that b* ***is within distance*** *D* ***of*** *a.*

Problem 18. *You have previously used Python to list the first hundred terms of the sequence $\left(\frac{n+1}{2n}\right)$. Now, sketch the graph of the function $f : \mathbb{N} \to \mathbb{R}; f(n) = \frac{n+1}{2n}$.*

Write out the 1-tail, 10-tail and 100-tail of the sequence.

Based on your calculations and your graph, can you guess at a number L that the sequence "tends to"?

Problem 19. *For the sequence and guess for L in the problem above, write a program that takes any positive number D as an input and outputs some natural number k for which $\frac{k+1}{2k}$ is within distance D of L.*

Problem 20. *Prove that no matter how small D is in the problem above, some value k exists such that $\frac{k+1}{2k}, \frac{k+2}{2(k+1)}, \frac{k+3}{2(k+2)}$, etc., are all within distance D of L.*

Interpret your finding geometrically, using the concept of k-tails.

Problem 21. *Repeat the two exercises above for the sequences $\left(\frac{n-1}{2n}\right)$ and $(1/n)$.*

Problem 22. *Repeat the same exercises for the sequences $(U_n(f)), (L_n(f))$ and $(U_n(f) - L_n(f))$ for the function in Example 4 of Sheet 1.*

Problem 23. *Repeat the same exercises for the function in Example 10 of Sheet 1.*

Problem 24. *Does it make sense to repeat any of the exercises for the function in Example 13 of Sheet 1? Do so whenever possible.*

Problem 25. *Does it make sense to repeat any of the exercises above for the function in Example 11 of Sheet 1? Do so whenever possible.*

Question 26. [1] *Using the concept of k-tails (or otherwise), come up with a definition of what it means for a sequence (a_n) to "tend to" L, and what it means for a sequence to not tend to anything at all.*

Sheet 3

Definition 27. *Given a sequence (a_n) in $\mathbb{R}$ and some real number L, we say that (a_n)* ***tends to*** *L if the following happens:*

No matter which real number $D > 0$ we choose, we can find some natural number k such that all terms in the k-tail of (a_n) are within distance D of L.

If no such L exists, we will say that (a_n) ***tends nowhere.***

Example 28. [2] *For the constant sequence $(2, -2, -2, \dots)$, pick $D = 0.5$ and find a k that satisfies the condition above. Do the same for $D = 0.00073$. Prove that the constant sequence $(a, a, a, \dots)$ tends to a for every real number a..*

Example 29. *For the sequence $(1/n)$, pick $D = 0.65$ and find a k that satisfies the condition in the definition above. Do the same for $D = 0.0014$. Then prove that the sequence $(1/n)$ tends to 0.*

Problem 30. *Come up with a few other types of sequences that tend to 0.*

Problem 31. *Come up with a few types of sequences that tend to $1/2$.*

Problem 32. *Complete the statement "A sequence (a_n)* ***does not*** *tend to L if ____*
____".

Problem 33. *Find a sequence that tends nowhere.*

Proposition 34. *A sequence of real numbers cannot tend to two different real numbers.*

Definition 35. *Let $f : [a, b] \to \mathbb{R}$ be a function. If the sequences $(U_n(f))$ and $(L_n(f))$ are both defined, and both tend to the same number L, then we say that f* ***is integrable*** *and write*

$$\int_a^b f(x)dx = L.$$

Example 36. *Show that the function in Example 4 is integrable.*

Example 37. *Show that the function in Example 9 is integrable.*

Problem 38. *Is the function from Example 13 in Sheet 1 integrable?*

Problem 39. *Is the function from Example 11 in Sheet 1 integrable?*

Question 40. *Can you think of a function $f : [a, b] \to \mathbb{R}$ for which $U_n(f)$ or $L_n(f)$ is not defined? How about one for which these are not defined, but which you believe should be integrable?*

Sheet 4

Definition 41. *Suppose $f : (a,b) \to \mathbb{R}$ is a function and $c,d \in (a,b)$. The* ***secant line of f between c and d****, $L_{c,d}$, is the straight line passing through $(c, f(c))$ and $(d, f(d))$.*

Example 42. *For each of the functions and points c given below, sketch the secant line from c to various points d in the domain. Calculate the slope of each secant line. What do you observe about the slopes of the secant lines for points d that are close to c?*

i) $f : (0,1) \to \mathbb{R}; f(x) = 7$; $c = 0.5$.

ii) $g : (-1,2) \to \mathbb{R}; g(x) = x$; $c = 0.5$.

iii) $h : (0,1) \to \mathbb{R}; h(x) = x^2$; $c = 0.5$.

iv) $k : (-1,1) \to \mathbb{R}; k(x) = |x|$; $c = 0$.

v) $p : (-1,1) \to \mathbb{R}; p(x) = \begin{cases} 3 & \text{for } -1 < x \leq 0 \\ -2 & \text{for } 0 < x < 1 \end{cases}; \qquad c = 0.$

Problem 43. *Let $f : (a,b) \to \mathbb{R}$ be a function and $c \in (a,b)$. Come up with a definition of what it means for f to be differentiable at c with derivative m based on the following idea:*

f is differentiable at c with derivative m if the slopes of the secant lines $L_{c,d}$ "tend to" m as the d's "tend to" c.

Check carefully that your definition implies all of the following:

i) *f from Example 42 is differentiable at* 0.5 *with derivative* 0.

ii) *g from Example 42 is differentiable at* 0.5 *with derivative* 1.

iii) *h from Example 42 is differentiable at* 0.5 *with derivative* 1.

iv) *k from Example 42 is not differentiable at* 0.

v) *p from Example 42 is not differentiable at* 0.

Chapter 2

Delving deeper into sequences

Sheet 5, Part 1

In the definitions below, (a_n) is a sequence in some $A \subseteq \mathbb{R}$.

Definition 44. *Let $L \in A$. We say that (a_n)* ***tends to*** *L* ***in*** *A no matter which real number $D > 0$ we take, we can find some natural number k such that all terms in the k-tail of (a_n) are within distance D of L.*

Definition 45. *We say (a_n) is* ***bounded*** *if there is some $B > 0$ such that $|a_n| < B$ for each natural number n.*

Definition 46. [3] *a) We say (a_n) is* ***monotonically increasing*** *if, for each natural number n, $a_n \leq a_{n+1}$.*

b) We say (a_n) is ***monotonically decreasing*** *if, for each natural number n, $a_n \geq a_{n+1}$.*

c) A ***monotonic sequence*** *is a sequence that is either monotonically increasing or monotonically decreasing.*

Example 47. *Give some examples of monotonic sequences, and verify that they satisfy Definition 46.*

Problem 48. *Write out what it means for a sequence to* **not** *be monotonic. Use this to give a few examples of non-monotonic sequences.*

Example 49. *Give some examples of bounded sequences, and verify that they satisfy Definition 45.*

Problem 50. *Write out what it means for a sequence to* **not** *be bounded. Use this to give a few examples of non-bounded sequences.*

Problem 51. [4] *Fill in the table below, considering all possible combinations of the three given properties, and give examples of sequences wherever possible. For which rows do you think it's impossible to find an example?*

	Tends somewhere	*Bounded*	*Monotonic*	*Example (if possible)*	*$A \subseteq \mathbb{R}$ in which the sequence lies*
1	*Yes*	*Yes*	*Yes*		
2	*Yes*	*Yes*	*No*		
3					
4					
5					
6					
7					
8					

Sheet 5, Part 2

Proposition 52. *Suppose $A \subseteq \mathbb{R}$ and (a_n) tends to L in A. Then (a_n) is bounded.*

Definition 53. *Let $f : \mathbb{N} \to \mathbb{R}$ be a sequence. Let $g : \mathbb{N} \to \mathbb{N}$ be a strictly increasing function. Then $f \circ g : \mathbb{N} \to \mathbb{R}$ is called a **subsequence** of f.*

Example 54. *Consider the sequence $((-1)^n)$. Show that $(1,1,1,\dots)$ and $(-1,-1,-1,\dots)$ are both subsequences. Is $(1,1,-1,1,1,-1,\dots)$ also a subsequence?*

Example 55. *Find an unbounded sequence in the reals for which some subsequences tend nowhere and others don't.*

Problem 56. *Explain why we can also think of a subsequence of $(a_n)_{n=1}^{\infty}$ as $(a_{n_i})_{i=1}^{\infty}$ for some infinite sequence $n_1 < n_2 < \dots$ of natural numbers.*

Example 57. *Show that every k-tail of a sequence is also a subsequence.*

Lemma 58. [5] *Every sequence has a monotonic subsequence.*

Chapter 3

The structure of the reals

Sheet 6 [6]

Axiom 59. *The set of real numbers is a **field** under addition and multiplication, i.e.,*

i) $a+b=b+a$ *for all* $a,b\in\mathbb{R}$

ii) $a+(b+c)=(a+b)+c$ *for all* $a,b,c\in\mathbb{R}$

iii) *There is a real number* 0 *such that* $a+0=a$ *for every* $a\in\mathbb{R}$

iv) *For every* $a\in\mathbb{R}$*, there is a number* $-a\in\mathbb{R}$ *such that* $a+(-a)=0$

v) $ab=ba$ *for all* $a,b\in\mathbb{R}$

vi) $a(bc)=(ab)c$ *for all* $a,b,c\in\mathbb{R}$

vii) *There is a real number* 1 *such that* $1a=a$ *for every* $a\in\mathbb{R}$

viii) *If* $a\neq 0$*, there is a real number* a^{-1} *such that* $aa^{-1}=1$

ix) $a(b+c)=ab+ac$ *for all* $a,b,c\in\mathbb{R}$

Axiom 60. *The field of real numbers is an **ordered field**, i.e., for all* $a,b,c,\in\mathbb{R}$*,*

i) *If* $a<b$*, then* $a+c<b+c$

ii) *If* $a>0,b>0$*, then* $ab>0$

Axiom 61. *Every bounded monotonic sequence of real numbers tends somewhere.*

Example 62. *Are* $\mathbb{N}$ *and* $\mathbb{Z}$ *ordered fields under the usual definitions of addition, multiplication and order? Exactly which parts of the conditions hold?*

Example 63. *Is $\mathbb{Q}$ an ordered field under the usual definitions of addition, multiplication and order?*

Example 64. *Is the set of irrationals an ordered field under the usual definitions of addition, multiplication and order?*

Problem 65. *Does Axiom 3 hold in the set of rational numbers? How about in the set of irrational numbers? Find a few subsets of the reals where you believe Axiom 3 holds, and some where it doesn't.*

Proposition 66. *Suppose x, y are real numbers.*

a) The product of -1 and x is the additive inverse of x from Axiom 1 iv).

b) If $x > 0$ and $y < 0$, then $xy < 0$.

Proposition 67. *Suppose x, y, z are real numbers and $x < y$.*

a) If $z > 0$, then $xz < yz$.

b) If $z < 0$, then $xz > yz$.

Proposition 68. [7] *Given any real $x > 0$, there is some natural number larger than x.*

Corollary 69. *For any real number x, both the smallest integer greater than or equal to x and the largest integer less than or equal to x exist. These are denoted by $\lceil x \rceil$ and $\lfloor x \rfloor$ respectively.*

Proposition 70. *Given any two real numbers $ax < y$, there is a rational number q such that $x < q < y$.*

Theorem 71. *Every bounded sequence of real numbers has a subsequence that tends somewhere.*

Chapter 4

Continuity

Sheet 7

Definition 72. *Let $A \subseteq \mathbb{R}$, and let $a \in A$. Let $f : A \to \mathbb{R}$ be a function. We say that f is* ***continuous at*** *a if given any sequence (a_n) tending to a in A, the sequence $(f(a_n))$ tends to $f(a)$. If f is continuous at all points on its domain, we say f is* ***continuous****.*

Problem 73. *Write out what it means for a function to* ***not*** *be continuous at a point. What about for a function to not be continuous?*

Example 74. *Fix any $c \in \mathbb{R}$. Show that $f : \mathbb{R} \to \mathbb{R}; f(x) = c$ is continuous.*

Example 75. *Show that $g : \mathbb{R} \to \mathbb{R}; g(x) = x$ is continuous.*

Example 76. *Sketch the graph of the function $h : [0,1] \cup [2,3] \to \mathbb{R}; h(x) = x$. Is h continuous?*

Example 77. *Show that the function from Example 11 is not continuous. What happens if we change the domain to $[-1,0)$? How about if we change the domain to $[-1,0) \cup (0,1]$?*

Example 78. *Show that the function from Example 13 is not continuous.*

Example 79. *Consider the functions $f, g : (-\pi/2, \pi/2) \to \mathbb{R}$ given by $f(x) = \sin(x)$ and $g(x) = \tan(x)$. Pick a few sequences (a_n) in the domain tending to 0 in different ways. Use Python to examine the behaviour of the sequences $(\sin(a_n))$ and $(\tan(a_n))$. What do you observe? What does this tell you about the continuity of f and g?*

Example 80. *Sketch the graph of the function*

$$h : \mathbb{R} \to \mathbb{R}; h(x) = \begin{cases} \sin(1/x) & \textit{if } x \neq 0 \\ 0 & \textit{if } x = 0 \end{cases}.$$

Pick a few sequences (a_n) tending to 0 in different ways. Use Python to examine the behaviour of the sequence $(h(a_n))$. What do you observe? What does this tell you about the continuity of h?

Proposition 81. *Let $A \subseteq B \subseteq \mathbb{R}$. If $f : B \to \mathbb{R}$ is continuous, then the restriction $f|_A : A \to \mathbb{R}$ is also continuous.*

Problem 82. *What are all the sequences that tend to 1 in $\mathbb{N}$? Can you find a function from $\mathbb{N}$ to $\mathbb{R}$ that is not continuous at 1? Can you find a discontinuous function from $\mathbb{N}$ to $\mathbb{R}$?*

Problem 83. *Suppose $f : [0,1] \cap \mathbb{Q} \to \mathbb{R}$ is a continuous function for which $f(0) > 0$ and $f(1) < 0$. Must there exist some $c \in (0,1)$ for which $f(c) = 0$?*

Theorem 84. [8] *Suppose $f : [0,1] \to \mathbb{R}$ is a continuous function for which $f(0) > 0$ and $f(1) < 0$. Then there exists some $c \in (0,1)$ such that $f(c) = 0$.*

Sheet 8

Example 85. *Prove that* $\left(\frac{1}{2^{n-1}}\right)$ *tends to* 0.

Problem 86. *Suppose* (a_n) *and* (b_n) *are sequences tending to* A *and* B *respectively such that* $|a_n - b_n| \leq 1/2^n$ *for each* n*. Prove that* $A = B$.

Problem 87. *Roughly sketch the graph of the function*

$$f : [0,1] \to \mathbb{R}; f(x) = -x^5 - 12x^3 - 12x^2 + 20.$$

Check that $f(0) > 0$ *and* $f(1) < 0$*. We will use the algorithm below to approximate where the function crosses the* x *axis.*

- *Set* $a_1 = 0, b_1 = 1$*. Set* $M_1 = 1/2$.
- *Is* $f(M_1)$ *positive, negative or zero? If it is zero, we have found a point at which* f *crosses the x-axis, and can stop the algorithm. If it is positive, set* $a_2 = M_1, b_2 = b_1$*, and consider* $M_2 = 3/4$*. Otherwise, set* $a_2 = a_1, b_2 = M_1$*, and consider* $M_2 = 1/4$*. Verify that in either case,* $|a_2 - b_2| = 1/2$.
- *Is* $f(M_2)$ *positive, negative or zero? If it is zero, we can terminate the algorithm. Otherwise, choose* $a_3 = a_2$ *or* M_2 *and* $b_3 = b_2$ *or* M_2 *following the pattern that* $f(a_3) > 0, f(b_3) < 0$ *and* $|a_3 - b_3| = 1/4$*. Then fix* M_3 *as the midpoint between* a_3 *and* b_3.
- *Continue this process to select* a_n, b_n.

Explain the pattern for choosing a_n *and* b_n*. For each* $0 \leq n \leq 4$*, calculate* $a_n, b_n, f(a_n), f(b_n)$*, and mark them on the graph. Make a guess about the values of* $f(a_n)$ *and* $f(b_n)$ *as* n *gets larger.*

Problem 88. *For the function and the algorithm in the Problem 87, use Python to calculate the first* 25 *terms of* $(a_n), (b_n), (f(a_n))$ *and* $(f(b_n))$*. What do you observe?*

Example 89. *Repeat Problem 88 for the function* $g : [0,1] \to \mathbb{R}; g(x) = \cos(e^x)$.

Problem 90. *Prove Theorem 84 using the ideas in this worksheet.*

Theorem 91. [9] *Suppose* $f : [a,b] \to \mathbb{R}$ *is a continuous function for which* $f(a)f(b) < 0$*. Then there exists some* $c \in (a,b)$ *such that* $f(c) = 0$.

Chapter 5

Further explorations

Sheet 9 [10]

Question 92. *In the proof of Theorems 84 and 91 we kept taking smaller and smaller closed intervals inside a given closed interval until we were left with a single point.*

Consider a sequence of intervals of the form $[a_1,b_1],[a_2,b_2],[a_3,b_3],\ldots,$ *where each* $[a_{i+1},b_{i+1}]$ *is a subset of* $[a_i,b_i]$*. Explain why the intersection of all these intervals together is not empty. What conditions can you add to make sure that the intersection of all the intervals consists of a single point?*

Would the same be true for intervals of the form $[a_1,b_1),[a_2,b_2),[a_3,b_3),$ $\ldots$*? Or* $(a_1,b_1),(a_2,b_2),(a_3,b_3),\ldots$*?*

Question 93. *An important feature of the reals is that the square roots of non-negative real numbers are real numbers. Suppose a is a positive integer. Can you come up with a sequence of rational numbers that converges to* $\sqrt{a}$*?*

Question 94. *In Problem 82, you probably observed that every function from the natural numbers to the reals is continuous. Another way to state this is that the restriction of any function* $f:\mathbb{R}\to\mathbb{R}$ *to the natural numbers is continuous. Is this true if we replace the natural numbers by the rationals?*

Is the restriction of f *to a subset being continuous the same as* f *being continuous at every element in that subset? Can* f *be continuous at every rational but discontinuous? Can* f *be continuous at every rational number and discontinuous at every irrational number?*

Question 95. *Suppose* $f:[a,b]\to\mathbb{R}$ *is a continuous function with the following property (that we saw in Theorem 91): no matter what* p,q *we take in* $[a,b]$*, if* $f(p)f(q)<0$*, then there exists some* c *lying in between* p *and* q *such that* $f(c)=0$*. Show with an example that a function satisfying this property need not be continuous. Can you add further condition to ensure that a function satisfying this property is continuous?*

Question 96. *We chose Axiom 3 based on an important structural difference between the rationals and the reals: the reals have no gaps. There are other choices we could have made here, though.*

*A statement P is said to be **equivalent to** Axiom 3 if P implies Axiom 3 and Axiom 3 implies P. Come up with some statements (results from this problem set or statements of your own) that are equivalent to Axiom 3.*

Notes to the Instructor

[1]The ideas that come up here will hopefully motivate the definition in the next sheet.

[2]It seems like many people's intuition tells them that convergence should involve getting closer and closer to a limit, but never quite reaching it. One of many motivations for introducing some version of integration before sequences is that it is natural to everyone that the constant function should be integrable.

[3]Strictly increasing/decreasing sequences are not defined, but the idea is likely to come up in discussion and can be then mentioned by the instructor.

[4]The discussion following this problem can be quite rich. Some students might state that they cannot find bounded monotonic sequences that don't converge. Others may observe that sequences can be defined in some specific subsets of the reals and find an example in the rationals. Both these observations become important in the next worksheet.

[5]This is the first hard result in the course. If it takes the class a long time to come up with a proof, it's okay to assume the result and move on to the next worksheet, and encourage students to keep working on the proof on the side. A proof should be presented by someone before Theorem 71.

[6]There is a sudden jump here, which needs to be smoothened by a lecture or discussion in which the difference in structure between the reals and rationals observed in Sheet 5 is discussed, along with an introduction to axiomatic systems.

[7]This and the next few results are facts about the reals that are quite likely assumed by students and used in the previous worksheets. The instructor can add more questions here based on the approaches taken by their students.

[8]If someone manages to prove this theorem directly, the class can move on to the final worksheet. If not, Sheet 8 can be handed out after the class spends some time thinking about proof strategies, possibly modified based on the direction students want to try taking.

[9]Following the proof of this result, the instructor can give some glimpses into its importance, and some of its applications.

[10]The questions in this worksheet have less structure than the previous ones. They are meant for somewhat more open-ended explorations at the of the semester, or for interested students to try out after the course ends.

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